

Mohibul Hassan Chowdhury

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Professional Summary

Senior Software Engineer II with 7+ years building high-reliability SaaS and distributed backend systems. Drive architecture for a CPaaS/contact-center platform serving 24,000+ users at 99.99% uptime. Architected production Google Vertex AI/Gemini call-intelligence capabilities and led database, reliability, telephony, and payment initiatives that reduced production risk and infrastructure cost. Strong in system design, production debugging, and technical leadership.

Professional Experience

- **Cefalo (Offshore branch of Cefalo AS, Norway)** Dhaka, Bangladesh
Senior Software Engineer II Sep 2022 - Present
 - **Production AI Call Transcription:** Architected and shipped Google Vertex AI/Gemini transcription for call recordings with single and batch processing, multilingual prompting, credential management, confidence-based evaluation, resilient failure handling across backend and administration workflows, provider-reliability evaluation, and bring-your-own-key pricing trade-offs.
 - **Conversation Intelligence:** Designed transcript-derived summaries across backend, database, administration, and agent workflows, including revision-aware staleness, append-only lifecycle state, batch processing, retries, feature flags, and notifications.
 - **Database Cleanup & Migration:** Analyzed approximately 68M records and cross-system dependencies; led a phased migration with validation and rollback safeguards that removed 8 write triggers and 4 unused tables, eliminated approximately 1.8M unnecessary writes/month, reclaimed an estimated 50 to 100 GB on affected database instances, and contributed to an estimated USD 600 to 1,200/year in infrastructure savings.
 - **Subscription & Payments:** Built subscription and payment capabilities across frontend and backend, including new/lost/net analytics, payment forecasts, search and filtering, and lifecycle operations; later reviewed architecture for a new payment platform intended to expand customer buying and payment options.
 - **Telephony & Production Reliability:** Serve as a subject-matter expert for telephony and production reliability, tracing complex failures across Java services, Erlang state machines, SQL, anonymization, transcription, dialing, and messaging workflows to resolve correctness, privacy, and reliability issues.
 - **Technical Leadership:** Provide technical leadership through architecture and code reviews across Java, Erlang, AI, messaging, payment, and telephony systems; identify failure-mode, API-boundary, correctness, and scalability risks before merge and guide engineers toward simpler, safer designs.
- **Brain Station 23** Dhaka, Bangladesh
Software Engineer Dec 2020 - Sept 2022
 - **Backend System Design:** Built production-grade Spring Boot REST APIs using Spring Data Java Persistence API (JPA) / Hibernate with strong domain modeling and transactional integrity
 - **Multi-Tenant Security Architecture:** Designed and implemented custom authentication and authorization system supporting hierarchical roles across tenant boundaries
 - **Payment Systems Integration:** Integrated Stripe billing workflows (product catalog, transactions, provisioning) ensuring accurate financial flows
 - **Quality Engineering:** Expanded automated test coverage (JUnit, Mockito), reducing production defects and accelerating delivery velocity
 - **Enterprise Platform Contribution:** Contributed to AbbviePro multi-tenant healthcare platform (Adobe Experience Manager) with Java backend and AngularJS frontend
- **Wellddev** Dhaka, Bangladesh
Junior Software Engineer Sept 2019 - Nov 2020
 - **Frontend Engineering:** Built responsive chatbot interface using Vue.js/Nuxt.js with structured state management (Vuex/VuexORM)
 - **Backend API Development:** Developed RESTful services using Spring Boot and PostgreSQL
 - **UI System Design:** Built Storybook component library to standardize UI and accelerate development
 - **DevOps Foundations:** Dockerized development environments to ensure reproducible builds and reduce onboarding friction

Technical Skills

Languages	Java, SQL, Erlang, Python, TypeScript, JavaScript
Backend & APIs	Spring Boot, Spring Security, Spring AI, Jersey (JAX-RS), Spring Data JPA/Hibernate, JDBC, REST APIs
Distributed Systems	System Design, Microservices, High Availability, Data Modeling, Idempotency, Consistent Hashing
AI/LLM	Google Vertex AI, Gemini, AWS Bedrock, RAG, PGVector, LLM Integration, Speech-to-Text, Structured Output, Batch AI Pipelines
Data & Performance	MySQL, SQL Optimization, Database Migration, JMH, JMX, VisualVM, Eclipse MAT, Heap Analysis, GC Tuning
Cloud & DevOps	AWS (EKS, Bedrock), Docker, GitHub Actions, CI/CD
Telephony	WebRTC, SIP/PJSIP, Asterisk, Contact-Center/CCaaS/CPaaS Systems
Security & Testing	TLS/JSSE, OAuth2, JWT, JUnit, Mockito, Integration Testing, Selenium
Frontend	Vue.js, Nuxt.js, TypeScript, AngularJS

Selected Technical Projects & Expertise

- **AI/LLM Systems:** Designed Spring AI agent architectures with streaming chat, JDBC-backed persistent memory, Retrieval-Augmented Generation (RAG) with PGVector, tool invocation, AWS Bedrock/Google Vertex AI orchestration, and EKS deployment.
- **JVM Performance & Security:** Applied JMH, JMX, VisualVM, Eclipse MAT, and garbage-collection tuning for performance analysis; implemented TLS/JSSE, OAuth2/JWT authentication, and secure sandboxing patterns.
- **Distributed Systems:** Applied idempotent API patterns, consistent hashing, microservices decomposition, and high-availability design across Java and Erlang systems.

Technical Writing & Independent Work

- Baeldung contributor and creator of 100DaysOfJava / Java Learning Atlas, publishing practical material on JVM internals, performance, concurrency, security, distributed systems, and Spring AI/RAG.

Education

- **BRAC University** Dhaka, Bangladesh
Bachelor of Science in Computer Science 2019